

Ethan Kim

Contact Information

919 590-9528
ethankimufg@gmail.com
Chapel Hill, NC

Education

University of North Carolina at Chapel Hill
Bachelor of Science Computer Science
& Minor in Statistics and Operations
Expected 2029

Associates in Arts from Guilford Technical
Community College

Relevant Courses

Fall 2026:
COMP 210 - Analysis of Data Structures
COMP 283 - Discrete Structures
STOR 305 - Intro Decision Analytics
MATH 233 - Multi Vari Calc 1 (Calc 3)
APPL 101 - Intro to Engineering
Spring 2027:
COMP 211 - Systems Fundamentals
COMP 301 - Foundations of Programming
Comp 455 - Models of Languages and
Computation
MATH 347 - Linear Algebra for Applications
STOR 235 - Introduction to Probability

Skills

Languages

- Java
- Python
- JavaScript
- TypeScript
- HTML
- CSS
- Lua

Frameworks

- React
- Next.js
- FastAPI
- Flask
- Websockets
- SocketIO
- OpenCV
- YOLO

Robotics

- Kalman Filters for
Sensor Fusion
- Control Systems
- Mapping
- Motion Planning
- Inverse Kinematics
- Computer Vision

Hardware

- Jetson Orin Nano &
other Linux devices
- Arduino
- Realsense Depth
Cameras
- GPS & RTK GPS
- Radio
- Lidar

Certifications

Java Programming Certification from
Guilford Technical Community College



ethan-kim-590275394



DoctorKrill08

Portfolio



https://ethankimufgonline.com

Experience and Projects

North Carolina Agriculture and Technical State University CLAEM Internship - Intern (June - August 2026)

Helped PHD students create and implement **software** for **Outdoor
Agricultural mobile robots**.

- Developed **autonomous navigation** software for mobile robots
using **LiDAR**, **depth cameras**, and **RTK GPS**.
- Designed **Kalman filter** sensor fusion integrating RTK GPS, **IMU**,
and **wheeled odometry** for **outdoor localization**.
- **Combined LiDar** and **Depth Cameras perception** with
localization to create a **Dynamic Obstacle Map**.
- Implemented **Dynamic Window Approach** local path planning for
obstacle avoidance and navigation.
- Built a **Web interface** for robot telemetry and control with a
python fastAPI backend, uvicorn server, a React Next.js frontend,
and Websockets as the communication protocol.
- Trained and integrated **YOLO object detection** models for
robotic arm use.
- Developed **inverse kinematics algorithms** for robotic arm
positioning and movement.

Undertale Fighting Game - Creator and Programmer (March 2022 - Present)

- Created, designed and **programmed** a multiplayer Roblox game
with over **6 million visits** and over **2 million unique players**
- Designed gameplay, visual effects, movement, and interfaces.

First Tech Challenge Team 10195 Captain and Lead Programmer (May 2024 - June 2026)

- Led software development for a competitive First Tech Challenge
Robot.
- Implemented **PID Control** for Motor Control based on Encoder
and Camera data.
- Created a **Kalman Filters** for Sensor fusion between **Wheeled
Odometry**, **IMU**, and **April Tags**.
- As lead programmer created robust autonomous routines and
achieved North Carolina **Autonomous** Offensive Power Rating
State Record in 2024 - 2025.
- As captain led the team to **win their first competition** since 2017
in the 2025 - 2026 season.

Volunteering

YB Champions Taekwondo Instructor

- Led class and instructed students of all ages for over 3 years,
1000+ hours

ECG Robotics Programming Camp Host/Leader

- Hosted and led a 5 day long programming camp that taught
rookie members (from multiple schools, 7th to 10th grade) how
to code in Java and some basic robotics concepts.